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Discrete Time Control

In this chapter we discuss discrete time Riccati equations and some of their applications in discrete time control. A simpler form of a Riccati equation is the so-called Lyapunov equation. Hence we will start from the solutions of a discrete Lyapunov equation which are given in section 21.1. Section 21.2 presents the basic property of a Riccati equation solution as well as the necessary and sufficient conditions for the existence of a solution to the LQR problem related Riccati equation. Various different characterizations of a bounded real transfer matrix are presented in section 21.3. The key is the relationship between the existence of a solution to a Riccati equation and the norm bound of a stable transfer matrix. Section 21.4 collects some useful matrix function factorizations and characterizations. In particular, state space criteria are stated (mostly without proof) for a transfer matrix to be inner, for the existence of coprime factorizations, inner-outer factorizations, normalized coprime factorizations, and spectral factorizations. The discrete time $\mathcal{H}_2$ optimal control will be considered briefly in Section 21.5. Finally, the discrete time balanced model reduction is considered in section 21.6 and 21.7.

21.1 Discrete Lyapunov Equations

Let A , B , and Q be real matrices with appropriate dimensions, and consider the following linear equation:

$$AXB - X + Q = 0. \quad (21.1)$$

Lemma 21.1 *The equation (21.1) has a unique solution if and only if $\lambda_i(A)\lambda_j(B) \neq 1$ for all i, j .*

Proof. Analogous to the continuous case. $\square$

Remark 21.1 If $\lambda_i(A)\lambda_j(B) = 1$ for some i, j , then the equation (21.1) has either no solution or more than one solution depending on the specific data given. If $B = A^*$ and $Q = Q^*$, then the equation is called the discrete Lyapunov equation. $\heartsuit$

The following results are analogous to the corresponding continuous time cases, so they will be stated without proof.

Lemma 21.2 *Let Q be a symmetric matrix and consider the following Lyapunov equation:*

$$AXA^* - X + Q = 0$$

1. *Suppose that A is stable, and then the following statements hold:*

- (a) $X = \sum_{i=0}^{\infty} A^i Q (A^*)^i$ and $X \geq 0$ if $Q \geq 0$.
- (b) if $Q \geq 0$, then (Q, A) is observable iff $X > 0$.

2. *Suppose that X is the solution of the Lyapunov equation; then*

- (a) $|\lambda_i(A)| \leq 1$ if $X > 0$ and $Q \geq 0$.
- (b) A is stable if $X \geq 0$, $Q \geq 0$ and (Q, A) is detectable.

21.2 Discrete Riccati Equations

This section collects some basic results on the discrete Riccati equations. So the presentation of this section and the sections to follow will be very much like the corresponding sections in Chapter 13. Just as the continuous time Riccati equations play the essential roles in continuous $\mathcal{H}_2$ and $\mathcal{H}_\infty$ theories, the discrete time Riccati equations play the essential roles in discrete time $\mathcal{H}_2$ and $\mathcal{H}_\infty$ theories.

Let a matrix $S \in \mathbb{R}^{2n \times 2n}$ be partitioned into four $n \times n$ blocks as

$$S := \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix},$$

and let $J = \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix} \in \mathbb{R}^{2n \times 2n}$; then S is called *symplectic* if $J^{-1}S^*J = S^{-1}$. A symplectic matrix has no eigenvalues at the origin, and, furthermore, it is easy to see that if λ is an eigenvalue of a symplectic matrix S , then $\bar{\lambda}$, $1/\lambda$, and $1/\bar{\lambda}$ are also eigenvalues of S .

Let A , Q , and G be real $n \times n$ matrices with Q and G symmetric and A nonsingular. Define a $2n \times 2n$ matrix:

$$S := \begin{bmatrix} A + G(A^*)^{-1}Q & -G(A^*)^{-1} \\ -(A^*)^{-1}Q & (A^*)^{-1} \end{bmatrix}.$$

Then S is a symplectic matrix. Assume that S has no eigenvalues on the unit circle. Then it must have n eigenvalues in $|z| < 1$ and n in $|z| > 1$. Consider the two n -dimensional spectral subspaces $\mathcal{X}_-(S)$ and $\mathcal{X}_+(S)$: the former is the invariant subspace corresponding to eigenvalues in $|z| < 1$, and the latter corresponds to eigenvalues in $|z| > 1$. After finding a basis for $\mathcal{X}_-(S)$, stacking the basis vectors up to form a matrix, and partitioning the matrix, we get

$$\mathcal{X}_-(S) = \text{Im} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$$

where $T_1, T_2 \in \mathbb{R}^{n \times n}$. If T_1 is nonsingular or, equivalently, if the two subspaces

$$\mathcal{X}_-(S), \quad \text{Im} \begin{bmatrix} 0 \\ I \end{bmatrix}$$

are complementary, we can set $X := T_2 T_1^{-1}$. Then X is uniquely determined by S , i.e., $S \mapsto X$ is a function which will be denoted Ric ; thus, $X = Ric(S)$. As in the continuous time case, we make the following definition.

Definition 21.1 The domain of Ric , denoted by $\text{dom}(Ric)$, consists of all $(2n \times 2n)$ symplectic matrices S such that S has no eigenvalues on the unit circle and the two subspaces $\mathcal{X}_-(S)$ and $\text{Im} \begin{bmatrix} 0 \\ I \end{bmatrix}$ are complementary.

Theorem 21.3 Suppose $S \in \text{dom}(Ric)$ and $X = Ric(S)$. Then

- (a) X is unique and symmetric;
- (b) $I + XG$ is invertible and X satisfies the algebraic Riccati equation

$$A^*XA - X - A^*XG(I + XG)^{-1}XA + Q = 0; \quad (21.2)$$

- (c) $A - G(I + XG)^{-1}XA = (I + GX)^{-1}A$ is stable.

Note that the discrete Riccati equation in (21.2) can also be written as

$$A^*(I + XG)^{-1}XA - X + Q = 0.$$

Remark 21.2 In the case that A is singular, all results presented in this chapter will still be true if the eigenvalue problem of S is replaced by the following generalized eigenvalue problem:

$$\lambda \begin{bmatrix} I & G \\ 0 & A^* \end{bmatrix} - \begin{bmatrix} A & 0 \\ -Q & I \end{bmatrix}$$

and $\mathcal{X}_-(S)$ is taken to be the subspace spanned by the *generalized principal vectors* corresponding to those generalized eigenvalues in $|z| < 1$. Here the generalized principal

vectors corresponding to a generalized eigenvalue λ are referred to the set of vectors $\{x_1, \dots, x_k\}$ satisfying

$$\begin{aligned} \begin{bmatrix} A & 0 \\ -Q & I \end{bmatrix} x_1 &= \lambda \begin{bmatrix} I & G \\ 0 & A^* \end{bmatrix} x_1 \\ \left(\begin{bmatrix} A & 0 \\ -Q & I \end{bmatrix} - \lambda \begin{bmatrix} I & G \\ 0 & A^* \end{bmatrix} \right) x_i &= \begin{bmatrix} I & G \\ 0 & A^* \end{bmatrix} x_{i-1}, \quad i = 2, \dots, k. \end{aligned}$$

See Dooren [1981] and Arnold and Laub [1984] for details. ♡

Proof. (a): Since $S \in \text{dom}(\text{Ric})$, $\exists T = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$ such that

$$\mathcal{X}_-(S) = \text{Im} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$$

and T_1 is invertible. Let $X := T_2 T_1^{-1}$, then

$$\mathcal{X}_-(S) = \text{Im} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \text{Im} \begin{bmatrix} I \\ X \end{bmatrix} T_1 = \text{Im} \begin{bmatrix} I \\ X \end{bmatrix}.$$

Obviously, X is unique since

$$\text{Im} \begin{bmatrix} I \\ X_1 \end{bmatrix} = \text{Im} \begin{bmatrix} I \\ X_2 \end{bmatrix}$$

iff $X_1 = X_2$. Now let us show that X is symmetric. Since

$$XT_1 = T_2, \tag{21.3}$$

pre-multiply by T_1^* to get

$$T_1^* X T_1 = T_1^* T_2.$$

We only need to show that $T_1^* T_2$ is symmetric.

Since $\mathcal{X}_-(S)$ is a stable invariant subspace, there is a stable $n \times n$ matrix S_- such that

$$ST = TS_-. \tag{21.4}$$

Pre-multiply by $S_-^* T^* J$ and note that $S^* J S = J$; we get

$$(S_-^* T^* J) T S_- = (S_-^* T^* J) S T = (S_-^* T^*) J S T = T^* S^* J S T = T^* J T$$

$$S_-^* (T^* J T) S_- - T^* J T = 0.$$

This is a Lyapunov equation and S_- is stable. Hence there is a unique solution

$$T^*JT = 0$$

i.e.,

$$T_2^*T_1 = T_1^*T_2.$$

Thus we have $X = X^*$ since T_1 is nonsingular.

(b): To show that X is the solution of the Riccati equation, we note that equation (21.4) can be written as

$$S \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} T_1 S_- T_1^{-1}. \quad (21.5)$$

Pre-multiply (21.5) by $\begin{bmatrix} -X & I \end{bmatrix}$ to get

$$\begin{bmatrix} -X & I \end{bmatrix} S \begin{bmatrix} I \\ X \end{bmatrix} = 0.$$

Equivalently, we get

$$-XA + (I + XG)(A^*)^{-1}(X - Q) = 0. \quad (21.6)$$

We now show that $I + XG$ is invertible. Suppose $I + XG$ is not invertible, then $\exists v \in \mathbb{R}^n$ such that

$$v^*(I + XG) = 0. \quad (21.7)$$

Pre-multiply (21.6) by v^* , and we get $v^*XA = 0$. Since A is assumed to be invertible, it follows that $v^*X = 0$. This, in turn, implies $v = 0$ by (21.7). Hence $I + XG$ is invertible and the equation (21.6) can be written as

$$A^*(I + XG)^{-1}XA - X + Q = 0.$$

This is the Riccati equation (21.2).

(c): To show that X is the stabilizing solution, pre-multiply (21.5) by $\begin{bmatrix} I & 0 \end{bmatrix}$ to get

$$A + G(A^*)^{-1}Q - G(A^*)^{-1}X = T_1 S_- T_1^{-1}.$$

The above equation can be simplified by using Riccati equation to get

$$(I + GX)^{-1}A = T_1 S_- T_1^{-1}$$

which is stable. □

Remark 21.3 Let $\mathcal{X}_+(S) = \text{Im} \begin{bmatrix} \tilde{T}_1 \\ \tilde{T}_2 \end{bmatrix}$ and suppose that $\tilde{T}_1$ is nonsingular. Then the Riccati equation has an anti-stabilizing solution $\tilde{X} = \tilde{T}_2 \tilde{T}_1^{-1}$ such that $(I + G\tilde{X})^{-1}A$ is antistable. $\heartsuit$

Lemma 21.4 Suppose that G and Q are positive semi-definite and that S has no eigenvalues on unit circle. Let $\mathcal{X}_-(S) = \text{Im} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$; then $T_1^* T_2 \geq 0$.

Proof. Let $T = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}$ and S_- be such that

$$ST = TS_-, \quad (21.8)$$

and S_- has all eigenvalues inside of the unit disc. Let

$$U_k = TS_-^k, \quad k = 0, 1, \dots$$

Then $U_{k+1} = SU_k$ with $U_0 = T$. Defining $V = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix}$, we get $T_1^* T_2 = U_0^* V U_0$.

Further define

$$\begin{aligned} Y_k &:= -U_k^* V U_k + U_0^* V U_0 \\ &= -\sum_{i=0}^{k-1} (U_{i+1}^* V U_{i+1} - U_i^* V U_i) \\ &= -\sum_{i=0}^{k-1} U_i^* (S^* V S - V) U_i. \end{aligned}$$

Now

$$S^* V S - V = \begin{bmatrix} I & -Q \\ 0 & I \end{bmatrix} \begin{bmatrix} -Q & 0 \\ 0 & -A^{-1}G(A^*)^{-1} \end{bmatrix} \begin{bmatrix} I & 0 \\ Q & I \end{bmatrix} \leq 0$$

since G and Q are assumed to be positive semi-definite. So $Y_k \geq 0$ for all $k \geq 0$. Note that $U_k \rightarrow 0$ as $k \rightarrow \infty$ since S_- has all eigenvalues inside the unit disc. Therefore $T_1^* T_2 = \lim_{k \rightarrow \infty} Y_k \geq 0$. $\square$

Lemma 21.5 Suppose that G and Q are positive semi-definite. Then $S \in \text{dom}(\text{Ric})$ iff (A, G) is stabilizable and S has no eigenvalues on the unit circle.

Proof. The necessary part is obvious. We now show that the stabilizability of (A, G) and S having no eigenvalues on the unit circle are, in fact, sufficient. To show this, we only need to show that T_1 is nonsingular, i.e. $\text{Ker } T_1 = 0$. First, it is claimed that $\text{Ker } T_1$ is S_- -invariant. To prove this, let $x \in \text{Ker } T_1$. Rewrite (21.8) as

$$(A + G(A^*)^{-1}Q)T_1 - G(A^*)^{-1}T_2 = T_1 S_- \quad (21.9)$$

$$-(A^*)^{-1}QT_1 + (A^*)^{-1}T_2 = T_2 S_- \quad (21.10)$$

Substitute (21.10) into (21.9) to get

$$AT_1 - GT_2 S_- = T_1 S_- \quad (21.11)$$

and pre-multiply the above equation by $x^* S_-^* T_2^*$ and post-multiply x to get

$$-x^* S_-^* T_2^* GT_2 S_- x = x^* S_-^* T_2^* T_1 S_- x.$$

According to Lemma 21.4 $T_2^* T_1 \geq 0$, we have

$$GT_2 S_- x = 0.$$

This in turn implies that $T_1 S_- x = 0$ from (21.11). Hence $\text{Ker } T_1$ is invariant under S_- .

Now to prove that T_1 is nonsingular, suppose on the contrary that $\text{Ker } T_1 \neq 0$. Then $S_-|_{\text{Ker } T_1}$ has an eigenvalue, λ , and a corresponding eigenvector, x :

$$S_- x = \lambda x \quad (21.12)$$

$$|\lambda| < 1, \quad 0 \neq x \in \text{Ker } T_1.$$

Post-multiply (21.10) by x to get

$$(A^*)^{-1}T_2 x = T_2 S_- x = \lambda T_2 x.$$

Now if $T_2 x \neq 0$, then $1/\bar{\lambda}$ is an eigenvalue of A and, furthermore, $0 = GT_2 S_- x = \lambda GT_2 x$, so $GT_2 x = 0$, which is contradictory to the stabilizability assumption of (A, G) . Hence

we must have $T_2 x = 0$. But this would imply $\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} x = 0$, which is impossible. $\square$

Lemma 21.6 *Let G and Q be positive semi-definite matrices and*

$$S = \begin{bmatrix} A + G(A^*)^{-1}Q & -G(A^*)^{-1} \\ -(A^*)^{-1}Q & (A^*)^{-1} \end{bmatrix}.$$

Then S has no eigenvalues on the unit circle iff (A, G) has no uncontrollable modes and (Q, A) has no unobservable modes on the unit circle.

Proof. ($\Leftarrow$) Suppose, on the contrary, that S has an eigenvalue $e^{j\theta}$. Then

$$S \begin{bmatrix} x \\ y \end{bmatrix} = e^{j\theta} \begin{bmatrix} x \\ y \end{bmatrix}$$

for some $\begin{bmatrix} x \\ y \end{bmatrix} \neq 0$, i.e.,

$$\begin{aligned} (A + G(A^*)^{-1}Q)x - G(A^*)^{-1}y &= e^{j\theta}x \\ -(A^*)^{-1}Qx + (A^*)^{-1}y &= e^{j\theta}y. \end{aligned}$$

Multiplying the second equation by G and adding it to the first one give

$$Ax - e^{j\theta}Gy = e^{j\theta}x \quad (21.13)$$

$$-Qx + y = e^{j\theta}A^*y. \quad (21.14)$$

Pre-multiplying equation (21.13) by $e^{-j\theta}y^*$ and equation (21.14) by x^* yield

$$\begin{aligned} e^{-j\theta}y^*Ax &= y^*Gy + y^*x \\ -x^*Qx + x^*y &= e^{j\theta}x^*A^*y. \end{aligned}$$

Thus

$$-y^*Gy - x^*Qx = 0.$$

It follows that

$$\begin{aligned} y^*G &= 0 \\ Qx &= 0. \end{aligned}$$

Substitute these relationships into (21.13) and (21.14) to get

$$\begin{aligned} Ax &= e^{j\theta}x \\ e^{j\theta}A^*y &= y. \end{aligned}$$

Since x and y cannot be zero simultaneously, $e^{j\theta}$ is either an unobservable mode of (Q, A) or an uncontrollable mode of (A, G) , a contradiction.

($\Rightarrow$): Suppose that S has no eigenvalue on the unit circle but $e^{j\theta}$ is an unobservable mode of (Q, A) and x is a corresponding eigenvector. Then it is easy to verify that

$$S \begin{bmatrix} x \\ 0 \end{bmatrix} = e^{j\theta} \begin{bmatrix} x \\ 0 \end{bmatrix},$$

so $e^{j\theta}$ is an eigenvalue of S , again a contradiction. The case for (A, G) having uncontrollable mode on the unit circle can be proven similarly. $\square$

Theorem 21.7 *Let G and Q be positive semi-definite matrices and*

$$S = \begin{bmatrix} A + G(A^*)^{-1}Q & -G(A^*)^{-1} \\ -(A^*)^{-1}Q & (A^*)^{-1} \end{bmatrix}.$$

Then $S \in \text{dom}(\text{Ric})$ iff (A, G) is stabilizable and (Q, A) has no unobservable modes on the unit circle. Furthermore, $X = \text{Ric}(S) \geq 0$ if $S \in \text{dom}(\text{Ric})$ and $X > 0$ if and only if (Q, A) has no unobservable stable modes.

Proof. Let $Q = C^*C$ for some matrix C . The first half of the theorem follows from Lemmas 21.5 and 21.6. Now rewrite the discrete Riccati equation as

$$A^*(I + XG)^{-1}X(I + GX)^{-1}A - X + A^*X(I + GX)^{-2}GXA + C^*C = 0 \quad (21.15)$$

and note that by definition $(I + GX)^{-1}A$ is stable and $A^*X(I + GX)^{-2}GXA + C^*C \geq 0$. Thus $X \geq 0$ by Lyapunov theorem. To show that the kernel of X has the refereed property, suppose $x \in \text{Ker}X$, pre-multiply (21.15) by x^* , and post-multiply by x to get

$$XAx = 0, \quad Cx = 0. \quad (21.16)$$

This implies that $\text{Ker}X$ is an A -invariant subspace. If $\text{Ker}X \neq 0$, then there is an $0 \neq x \in \text{Ker}X$, so $Cx = 0$, such that $Ax = \lambda x$. But for $x \in \text{Ker}X$, $(I + GX)^{-1}Ax = \lambda(I + GX)^{-1}x = \lambda x$, so $|\lambda| < 1$ since $(I + GX)^{-1}A$ is stable. Thus λ is an unobservable stable mode of (Q, A) .

On the other hand, suppose that $|\lambda| < 1$ is a stable unobservable mode of (Q, A) . Then there exists a $x \in \mathbb{C}^n$ such that $Ax = \lambda x$ and $Cx = 0$; do the same pre- and post-multiplications on (21.15) as before to get

$$|\lambda|^2 x^*(XG + I)^{-1}Xx - x^*Xx = 0.$$

This can be rewritten as

$$x^*X^{1/2}[|\lambda|^2(I + X^{1/2}GX^{1/2})^{-1} - I]X^{1/2}x = 0.$$

Now $X \geq 0$, $G \geq 0$, and $|\lambda| < 1$ imply that $|\lambda|^2(I + X^{1/2}GX^{1/2})^{-1} - I < 0$. Hence $Xx = 0$, i.e., X is singular. $\square$

Lemma 21.8 *Suppose that D has full column rank and let $R = D^*D > 0$; then the following statements are equivalent:*

- (i) $\begin{bmatrix} A - e^{j\theta}I & B \\ C & D \end{bmatrix}$ has full column rank for all $\theta \in [0, 2\pi]$.
- (ii) $((I - DR^{-1}D^*)C, A - BR^{-1}D^*C)$ has no unobservable modes on the unit circle, or, equivalently, $(D_\perp^*C, A - BR^{-1}D^*C)$ has no unobservable modes on the unit circle.

Proof. Suppose that $e^{j\theta}$ is an unobservable mode of $((I - DR^{-1}D^*)C, A - BR^{-1}D^*C)$; then there is an $x \neq 0$ such that

$$(A - BR^{-1}D^*C)x = e^{j\theta}x, \quad (I - DR^{-1}D^*)Cx = 0$$

i.e.,

$$\begin{bmatrix} A - e^{j\theta}I & B \\ C & D \end{bmatrix} \begin{bmatrix} I & 0 \\ -R^{-1}D^*C & I \end{bmatrix} \begin{bmatrix} x \\ 0 \end{bmatrix} = 0.$$

But this implies that

$$\begin{bmatrix} A - e^{j\theta}I & B \\ C & D \end{bmatrix} \quad (21.17)$$

does not have full column rank. Conversely, suppose that (21.17) does not have full column rank for some θ ; then there exists $\begin{bmatrix} u \\ v \end{bmatrix} \neq 0$ such that

$$\begin{bmatrix} A - e^{j\theta}I & B \\ C & D \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 0.$$

Now let

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} I & 0 \\ -R^{-1}D^*C & I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Then

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} I & 0 \\ R^{-1}D^*C & I \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} \neq 0$$

and

$$(A - BR^{-1}D^*C - e^{j\theta}I)x + By = 0 \quad (21.18)$$

$$(I - DR^{-1}D^*)Cx + Dy = 0. \quad (21.19)$$

Pre-multiply (21.19) by D^* to get $y = 0$. Then we have

$$(A - BR^{-1}D^*C)x = e^{j\theta}x, \quad (I - DR^{-1}D^*)Cx = 0$$

i.e., $e^{j\theta}$ is an unobservable mode of $((I - DR^{-1}D^*)C, A - BR^{-1}D^*C)$. $\square$

Corollary 21.9 Suppose that D has full column rank and denote $R = D^*D > 0$. Let S have the form

$$S = \begin{bmatrix} E + G(E^*)^{-1}Q & -G(E^*)^{-1} \\ -(E^*)^{-1}Q & (E^*)^{-1} \end{bmatrix}$$

where $E = A - BR^{-1}D^*C$, $G = BR^{-1}B^*$, $Q = C^*(I - DR^{-1}D^*)C$, and E is assumed to be invertible. Then $S \in \text{dom}(\text{Ric})$ iff (A, B) is stabilizable and $\begin{bmatrix} A - e^{j\theta}I & B \\ C & D \end{bmatrix}$ has full column rank for all $\theta \in [0, 2\pi]$. Furthermore, $X = \text{Ric}(S) \geq 0$.

Note that the Riccati equation corresponding to the symplectic matrix in Corollary 21.9 is

$$E^*XE - X - E^*XG(I + XG)^{-1}XE + Q = 0.$$

This equation can also be written as

$$A^*XA - X - (B^*XA + D^*C)^*(D^*D + B^*XB)^{-1}(B^*XA + D^*C) + C^*C = 0.$$

21.3 Bounded Real Functions

Let a real rational transfer matrix be given by

$$M(z) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$$

where again A is assumed to be nonsingular and the realization is assumed to have no uncontrollable and no unobservable modes on the unit circle. Note again that all results hold for A singular case with the same modification as in the last section. Define $M^\sim(z) := M^T(z^{-1})$. Then

$$M^\sim(z) = \left[\begin{array}{c|c} (A^*)^{-1} & -(A^*)^{-1}C^* \\ \hline B^*(A^*)^{-1} & D^* - B^*(A^*)^{-1}C^* \end{array} \right].$$

Lemma 21.10 Let $M(z) = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array} \right] \in \mathcal{RL}_\infty$ and let S be a symplectic matrix defined by

$$S := \left[\begin{array}{cc} A - BB^*(A^*)^{-1}C^*C & BB^*(A^*)^{-1} \\ -(A^*)^{-1}C^*C & (A^*)^{-1} \end{array} \right].$$

Then the following statements are equivalent:

- (i) $\|M(z)\|_\infty < 1$;
- (ii) S has no eigenvalues on the unit circle and $\|C(I - A)^{-1}B\| < 1$.

Proof. It is easy to compute that

$$[I - M^\sim(z)M(z)]^{-1} = \left[\begin{array}{cc|c} A - BB^*(A^*)^{-1}C^*C & -BB^*(A^*)^{-1} & B \\ (A^*)^{-1}C^*C & (A^*)^{-1} & 0 \\ \hline -B^*(A^*)^{-1}C^*C & -B^*(A^*)^{-1} & I \end{array} \right].$$

It is claimed that $[I - M^\sim(z)M(z)]^{-1}$ has no uncontrollable and/or unobservable modes on the unit circle.

To show that, suppose that $\lambda = e^{j\theta}$ is an uncontrollable mode of $[I - M^\sim(z)M(z)]^{-1}$. Then $\exists q = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \in \mathbb{C}^{2n}$ such that

$$q^* \begin{bmatrix} A - BB^*(A^*)^{-1}C^*C & -BB^*(A^*)^{-1} \\ (A^*)^{-1}C^*C & (A^*)^{-1} \end{bmatrix} = e^{j\theta} q^*, \quad q^* \begin{bmatrix} B \\ 0 \end{bmatrix} = 0.$$

Hence $q_1^*B = 0$ and

$$\begin{bmatrix} q_1^*A + q_2^*(A^*)^{-1}C^*C & q_2^*(A^*)^{-1} \end{bmatrix} = e^{j\theta} \begin{bmatrix} q_1^* & q_2^* \end{bmatrix}.$$

There are two possibilities:

1. $q_2 \neq 0$. Then we have $q_2^*(A^*)^{-1} = e^{j\theta} q_2^*$, i.e., $Aq_2 = e^{-j\theta} q_2$. This implies $e^{-j\theta}$ is an eigenvalue of A . This is a contradiction since $M(z) \in \mathcal{RL}_\infty$.
2. $q_2 = 0$. Then $q_1^*A = e^{j\theta} q_1^*$, which again implies that $M(z)$ has a mode on the unit circle if $q_1 \neq 0$, again a contradiction.

Similar proof can be done for observability, hence the claim is true.

Now note that

$$S = \begin{bmatrix} -I & \\ & I \end{bmatrix} \begin{bmatrix} A - BB^*(A^*)^{-1}C^*C & -BB^*(A^*)^{-1} \\ (A^*)^{-1}C^*C & (A^*)^{-1} \end{bmatrix} \begin{bmatrix} -I & \\ & I \end{bmatrix}.$$

Hence S does not have eigenvalues on the unit circle. It is clear that we have already proven that S has no eigenvalues on the unit circle iff $(I - M^\sim M)^{-1} \in \mathcal{RL}_\infty$. So it is sufficient to show that

$$\|M(z)\|_\infty < 1 \Leftrightarrow (I - M^\sim M)^{-1} \in \mathcal{RL}_\infty \text{ and } \|M(1)\| < 1.$$

It is obvious that the right hand side is necessary. To show that it is also sufficient, suppose $\|M(z)\|_\infty \geq 1$, then $\sigma_{max}(M(e^{j\theta})) = 1$ for some $\theta \in [0, 2\pi]$, since $\sigma_{max}(M(1)) < 1$ and $M(e^{j\theta})$ is continuous in θ . This implies that 1 is an eigenvalue of $M^*(e^{-j\theta})M(e^{j\theta})$, so $I - M^*(e^{-j\theta})M(e^{j\theta})$ is singular. This contradicts to $(I - M^\sim M)^{-1} \in \mathcal{RL}_\infty$. $\square$

In the above Lemma, we have assumed that the transfer matrix is strictly proper. We shall now see how to handle non-strictly proper case. For that purpose we shall focus our attention on the stable system, and we shall give an example below to show why this restriction is sometimes necessary for the technique to work.

We first note that $\mathcal{H}_\infty$ -norm of a stable system is defined as

$$\|M(z)\|_\infty = \sup_{|z| \geq 1} \bar{\sigma}(M(z)).$$

Then it is clear that $\|M(z)\|_\infty \geq \bar{\sigma}(M(\infty)) = \|D\|$. Thus in particular if $\|M(z)\|_\infty < 1$ then $I - D^*D > 0$.

On the other hand, if a function is only known to be in $\mathcal{RL}_\infty$, the above condition may not be true if the system is not stable.

Example 21.1 Let $0 < \alpha < 1/2$ and let

$$M_1(z) = \frac{z}{z - 1/\alpha} = \left[\begin{array}{c|c} 1/\alpha & \alpha \\ \hline 1 & 1 \end{array} \right] \in \mathcal{RL}_\infty.$$

Then $\|M_1(z)\|_\infty = \frac{\alpha}{1-\alpha} < 1$, but $1 - D^*D = 0$. In general, if $M \in \mathcal{RL}_\infty$ and $\|M\|_\infty < 1$, then $I - D^*D$ can be indefinite. $\diamond$

Lemma 21.11 Let $M(z) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty$. Then $\|M(z)\|_\infty < 1$ if and only if $N(z) \in \mathcal{RH}_\infty$ and $\|N(z)\|_\infty < 1$ where

$$N(z) = \left[\begin{array}{c|c} \frac{A + B(I - D^*D)^{-1}D^*C}{(I - DD^*)^{-1/2}C} & \frac{B(I - D^*D)^{-1/2}}{0} \\ \hline \frac{E}{\hat{C}} & \frac{\hat{B}}{0} \end{array} \right] =: \left[\begin{array}{c|c} E & \hat{B} \\ \hline \hat{C} & 0 \end{array} \right].$$

Proof. This is exactly the analogy of Corollary 17.4. $\square$

Theorem 21.12 Let $M(z) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty$ and define

$$\begin{aligned} E &:= A + B(I - D^*D)^{-1}D^*C \\ G &:= -B(I - D^*D)^{-1}B^* \\ Q &:= C^*(I - DD^*)^{-1}C. \end{aligned}$$

Suppose that E is nonsingular and define a symplectic matrix as

$$S := \left[\begin{array}{cc} E + G(E^*)^{-1}Q & -G(E^*)^{-1} \\ -(E^*)^{-1}Q & (E^*)^{-1} \end{array} \right].$$

Then the following statements are equivalent:

- (a) $\|M(z)\|_\infty < 1$;
- (b) S has no eigenvalues on the unit circle and $\|C(I - A)^{-1}B + D\| < 1$;
- (c) $\exists X \geq 0$ such that $I - D^*D - B^*XB > 0$ and

$$E^*XE - X - E^*XG(I + XG)^{-1}XE + Q = 0$$

and $(I + GX)^{-1}E$ is stable. Moreover, $X > 0$ if (C, A) is observable.

- (d) $\exists X > 0$ such that $I - D^*D - B^*XB > 0$ and

$$E^*XE - X - E^*XG(I + XG)^{-1}XE + Q < 0;$$

(e) $\exists X > 0$ such that

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^* \begin{bmatrix} X & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} - \begin{bmatrix} X & 0 \\ 0 & I \end{bmatrix} < 0;$$

(f) $\exists T$ nonsingular such that

$$\bar{\sigma} \left(\begin{bmatrix} TAT^{-1} & TB \\ CT^{-1} & D \end{bmatrix} \right) = \left\| \begin{bmatrix} T & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} T & 0 \\ 0 & I \end{bmatrix}^{-1} \right\| < 1;$$

(g) $\mu_{\Delta} \left(\begin{bmatrix} A & B \\ C & D \end{bmatrix} \right) < 1$ with $\Delta \in \mathbf{\Delta}$ and

$$\mathbf{\Delta} := \left\{ \begin{bmatrix} \delta_1 I_n & 0 \\ 0 & \Delta_2 \end{bmatrix} : \delta_1 \in \mathbb{C}, \Delta_2 \in \mathbb{C}^{m \times p} \right\} \subset \mathbb{C}^{(n+m) \times (n+p)}$$

(assuming that $M(s)$ is a $p \times m$ matrix and A is an $n \times n$ matrix).

Note that the Riccati equation in (c) can also be written as

$$A^*XA - X + (B^*XA + D^*C)^*(I - D^*D - B^*XB)^{-1}(B^*XA + D^*C) + C^*C = 0.$$

Proof. (a) $\Leftrightarrow$ (b) follows by Lemma 21.10

(a) $\Rightarrow$ (g) follows from Theorem 11.7.

(g) $\Rightarrow$ (f) follows from Theorem 11.5.

(f) $\Rightarrow$ (e) follows by letting $X = T^*T$.

(e) $\Rightarrow$ (d) follows by Schur complementary formula.

(d) $\Rightarrow$ (c) can be shown in the same way as in the proof of Theorem 13.11.

(c) $\Rightarrow$ (a) We shall only give the proof for $D = 0$ case, the case $D \neq 0$ can be transformed to the zero case by Lemma 21.11. Hence in the following we have $E = A$, $G = -BB^*$, and $Q = C^*C$.

Assuming (c) is satisfied by some $X \geq 0$ and considering the obvious relation with $z := e^{j\theta}$

$$\begin{aligned} (z^{-1}I - A^*)X(zI - A) + (z^{-1}I - A^*)XA + A^*X(zI - A) &= X - A^*XA \\ &= C^*C + A^*XB(I - B^*XB)^{-1}B^*XA. \end{aligned}$$

The last equality is obtained from substituting in Riccati equation. Now pre-multiply the above equation by $B^*(z^{-1}I - A)^{-1}$ and post-multiply by $(zI - A)^{-1}B$ to get

$$I - M^*(z^{-1})M(z) = W^*(z^{-1})W(z)$$

where

$$W(z) = \left[\begin{array}{c|c} A & B \\ \hline (I - B^*XB)^{-1/2}B^*XA & -(I - B^*XB)^{1/2} \end{array} \right].$$

Suppose $W(e^{j\theta})v = 0$ for some θ and v ; then $e^{j\theta}$ is a zero of $W(z)$ if $v \neq 0$. However, all the zeros of $W(z)$ are given by the eigenvalues of

$$A + B(I - B^*XB)^{-1}B^*XA = (I - BB^*X)^{-1}A$$

that are all inside of the unit circle. Hence $e^{j\theta}$ cannot be a zero of W .

Therefore, we get $I - M^*(e^{-j\theta})M(e^{j\theta}) > 0$ for all $\theta \in [0, 2\pi]$, i.e., $\|M\|_\infty < 1$. $\square$

The following more general results can be proven easily following the same procedure as in the proof of (c) $\Rightarrow$ (a).

Corollary 21.13 Let $M(z) = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array} \right] \in \mathcal{RL}_\infty$ and suppose $\exists X = X^*$ such that

$$A^*XA - X + A^*XB(I - B^*XB)^{-1}B^*XA + C^*C = 0.$$

Then

$$I - M^*(z^{-1})M(z) = W^*(z^{-1})(I - B^*XB)W(z)$$

where

$$W(z) = \left[\begin{array}{c|c} A & B \\ \hline (I - B^*XB)^{-1}B^*XA & -I \end{array} \right].$$

Moreover, the following statements hold:

- (1) if $I - B^*XB > 0 (< 0)$, then $\|M(z)\|_\infty \leq 1 (\geq 1)$;
- (2) if $I - B^*XB > 0 (< 0)$ and $|\lambda_i\{(I - BB^*X)^{-1}A\}| \neq 1$, then $\|M(z)\|_\infty < 1 (> 1)$.

Remark 21.4 As in the continuous time case, the equivalence between (a) and (b) in Theorem 21.12 can be used to compute the $\mathcal{H}_\infty$ norm of a discrete time transfer matrix.

♡

21.4 Matrix Factorizations

21.4.1 Inner Functions

A transfer matrix $N(z)$ is called *inner* if $N(z)$ is stable and $N^*(z)N(z) = I$ for all $z = e^{j\theta}$. Note that if N has no poles at the origin, then $N^*(e^{j\theta}) = N^\sim(e^{j\theta})$. A transfer matrix is called *outer* if all its transmission zeros are stable (i.e., inside of the unit disc).

Lemma 21.14 Let $N = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$ and suppose that $X = X^*$ satisfies

$$A^*XA - X + C^*C = 0.$$

Then

- (a) $D^*C + B^*XA = 0$ implies $N^\sim N = (D - CA^{-1}B)^*D$;
- (b) $X \geq 0$, (A, B) controllable, and $N^\sim N = (D - CA^{-1}B)^*D$ implies $D^*C + B^*XA = 0$.

Proof. The results follow by noting the following equality:

$$\begin{aligned} N^\sim(z)N(z) &= \left[\begin{array}{cc|c} A & 0 & B \\ \hline (A^*)^{-1}C^*C & (A^*)^{-1} & (A^*)^{-1}C^*D \\ \hline (D - CA^{-1}B)^*C & -B^*(A^*)^{-1} & (D - CA^{-1}B)^*D \end{array} \right] \\ &= \left[\begin{array}{cc|c} A & 0 & B \\ 0 & (A^*)^{-1} & XB + (A^*)^{-1}C^*D \\ \hline D^TC + B^*XA & -B^*(A^*)^{-1} & (D - CA^{-1}B)^*D \end{array} \right]. \end{aligned}$$

□

The following corollary is a special case of this lemma which gives the necessary and sufficient conditions of a discrete inner transfer matrix with the state-space representation.

Corollary 21.15 Suppose that $N(z) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty$ is a controllable realization; then $N(z)$ is inner if and only if there exists a matrix $X = X^* \geq 0$ such that

- (a) $A^*XA - X + C^*C = 0$
- (b) $D^*C + B^*XA = 0$
- (c) $(D - CA^{-1}B)^*D = D^*D + B^*XB = I$.

The following alternative characterization of the inner transfer matrix is often useful and insightful.

Corollary 21.16 *Let $N(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty$ and assume that there exists a T nonsingular such that*

$$P = \begin{bmatrix} TAT^{-1} & TB \\ CT^{-1} & D \end{bmatrix} \quad \text{and} \quad P^*P = I.$$

Then $N(z)$ is an inner. Furthermore, if the realization of N is minimal, then such T exists.

Proof. Rewrite $P^*P = I$ as

$$\begin{bmatrix} A^* & C^* \\ B^* & D^* \end{bmatrix} \begin{bmatrix} T^*T & \\ & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} T^*T & \\ & I \end{bmatrix}.$$

Let $X = T^*T$, and then

$$\begin{bmatrix} A^*XA - X + C^*C & A^*XB + C^*D \\ B^*XA + D^*C & B^*XB + D^*D - I \end{bmatrix} = 0.$$

This is the desired equation, so N is inner. On the other hand, if the realization is minimal, then $X > 0$. This implies that T exists. $\square$

In a similar manner, Corollary 21.15 can be used to derive the state-space representation of the complementary inner factor (CIF).

Lemma 21.17 *Suppose that a $p \times m$ ($p > m$) transfer matrix $N(z)$ (minimal) is inner; then there exists a $p \times (p - m)$ CIF $N_\perp \in \mathcal{RH}_\infty$ such that the matrix $\begin{bmatrix} N & N_\perp \end{bmatrix}$ is square and inner. A particular realization is*

$$N_\perp(z) = \left[\begin{array}{c|c} A & Y \\ \hline C & Z \end{array} \right]$$

where Y and Z satisfy

$$A^*XY + C^*Z = 0 \tag{21.20}$$

$$B^*XY + D^*Z = 0 \tag{21.21}$$

$$Z^*Z + Y^*XY = I. \tag{21.22}$$

Proof. Note that $\begin{bmatrix} N & N_\perp \end{bmatrix} = \left[\begin{array}{c|cc} A & B & Y \\ \hline C & D & Z \end{array} \right]$ is inner. Now it is easy to prove the results by using the formula in Corollary 21.15. $\square$

21.4.2 Coprime Factorizations

Recall that two transfer matrices $M(z), N(z) \in \mathcal{RH}_\infty$ are said to be right coprime if $\begin{bmatrix} M \\ N \end{bmatrix}$ is left invertible in $\mathcal{RH}_\infty$ i.e., $\exists U, V \in \mathcal{RH}_\infty$ such that

$$U(z)N(z) + V(z)M(z) = I.$$

The left coprime is defined analogously. A plant $G(z) \in \mathcal{RL}_\infty$ is said to have double coprime factorization if $\exists$ a right coprime factorization $G = NM^{-1}$, a left coprime factorization $G = \tilde{M}^{-1}\tilde{N}$, and $U, V, \tilde{U}, \tilde{V} \in \mathcal{RH}_\infty$ such that

$$\begin{bmatrix} V & U \\ -\tilde{N} & \tilde{M} \end{bmatrix} \begin{bmatrix} M & -\tilde{U} \\ N & \tilde{V} \end{bmatrix} = I. \quad (21.23)$$

The state space formulae for discrete time transfer matrix coprime factorization are the same as for the continuous time. They are given by the following theorem.

Theorem 21.18 Let $G(z) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RL}_\infty$ be a stabilizable and detectable realization. Choose F and L such that $A + BF$ and $A + LC$ are both stable. Let $U, V, \tilde{U}, \tilde{V}, N, M, \tilde{N}$, and $\tilde{M}$ be given as follows

$$\begin{aligned} \begin{bmatrix} M \\ N \end{bmatrix} &:= \left[\begin{array}{c|c} A + BF & BZ_r \\ \hline F & Z_r \\ \hline C + DF & DZ_r \end{array} \right] \\ \begin{bmatrix} \tilde{U} \\ \tilde{V} \end{bmatrix} &:= \left[\begin{array}{c|c} A + BF & LZ_l^{-1} \\ \hline F & 0 \\ \hline -(C + DF) & Z_l^{-1} \end{array} \right] \\ \begin{bmatrix} \tilde{M} \\ \tilde{N} \end{bmatrix} &:= \left[\begin{array}{c|cc} A + LC & L & B + LD \\ \hline Z_l C & Z_l & Z_l D \end{array} \right] \\ \begin{bmatrix} U \\ V \end{bmatrix} &:= \left[\begin{array}{c|cc} A + LC & L & -(B + LD) \\ \hline Z_l^{-1} F & 0 & Z_l^{-1} \end{array} \right] \end{aligned}$$

where Z_r and Z_l are any nonsingular matrices. Then $G = NM^{-1} = \tilde{M}^{-1}\tilde{N}$ are rcf and lcf, respectively, and (21.23) is satisfied.

Some coprime factorizations are particularly interesting, for example, the *coprime factorization with inner numerator*. This factorization in the case of $G(z) \in \mathcal{RH}_\infty$ yields an *inner-outer factorization*.

Theorem 21.19 *Assume that (A, B) is stabilizable, $\begin{bmatrix} A - e^{j\theta}I & B \\ C & D \end{bmatrix}$ has full column rank for all $\theta \in [0, 2\pi]$, and D has full column rank. Then there exists a right coprime factorization $G = NM^{-1}$ such that N is inner. Furthermore, a particular realization is given by*

$$\begin{bmatrix} M \\ N \end{bmatrix} := \left[\begin{array}{c|c} \frac{A + BF}{F} & \frac{BR^{-1/2}}{R^{-1/2}} \\ \hline C + DF & DR^{-1/2} \end{array} \right]$$

where

$$R = D^*D + B^*XB$$

$$F = -R^{-1}(B^*XA + D^*C)$$

and $X = X^* \geq 0$ is the unique stabilizing solution

$$A_D^*X(I + B(D^*D)^{-1}B^*X)^{-1}A_D - X + C^*D_\perp D_\perp^*C = 0$$

where $A_D := A - B(D^*D)^{-1}D^*C$.

Using Lemma 21.17, the complementary inner factor of N in Theorem 21.19 can be obtained as follows

$$N_\perp = \left[\begin{array}{c|c} \frac{A + BF}{C + DF} & \frac{Y}{Z} \end{array} \right]$$

where Y and Z satisfy

$$A^*XY + C^*Z = 0$$

$$B^*XY + D^*Z = 0$$

$$Z^*Z + Y^*XY = I.$$

Note that Y and Z are only related to F implicitly through X .

Remark 21.5 If $G(z) \in \mathcal{RH}_\infty$, then the denominator matrix M in Theorem 21.19 is an outer. Hence, the factorization $G = N(M^{-1})$ is an inner-outer factorization. $\heartsuit$

Suppose that the system $G(z)$ is not stable; then a *coprime factorization with inner denominator* can also be obtained by solving a special Riccati equation.

Theorem 21.20 Assume that $G(z) := \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RL}_\infty$ and that (A, B) is stabilizable. Then there exists a right coprime factorization $G = NM^{-1}$ such that M is inner if and only if G has no poles on the unit circle. A particular realization is

$$\left[\begin{array}{c} M \\ N \end{array} \right] := \left[\begin{array}{c|c} A + BF & BR^{-1/2} \\ \hline F & R^{-1/2} \\ C + DF & DR^{-1/2} \end{array} \right]$$

where

$$R = I + B^*XB$$

$$F = -R^{-1}B^*XA$$

and $X = X^* \geq 0$ is the unique stabilizing solution to

$$A^*X(I + BB^*X)^{-1}A - X = 0.$$

Another special coprime factorization is called *normalized coprime factorization* which has found applications in many control problems such as model reduction controller design and gap metric characterization.

Recall that a right coprime factorization of $G = NM^{-1}$ with $N, M \in \mathcal{RH}_\infty$ is called a *normalized right coprime factorization* if

$$M^*M + N^*N = I$$

i.e., if $\left[\begin{array}{c} M \\ N \end{array} \right]$ is an inner.

Similarly, an lcf $G = \tilde{M}^{-1}\tilde{N}$ is called a *normalized left coprime factorization* if $\left[\begin{array}{cc} \tilde{M} & \tilde{N} \end{array} \right]$ is a co-inner. Then the following results follow in the same way as for the continuous time case.

Theorem 21.21 Let a realization of G be given by

$$G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$$

and define

$$R = I + D^*D > 0, \quad \tilde{R} = I + DD^* > 0.$$

(a) Suppose that (A, B) is stabilizable and that (C, A) has no unobservable modes on the imaginary axis. Then there is a normalized right coprime factorization $G = NM^{-1}$

$$\left[\begin{array}{c} M \\ N \end{array} \right] := \left[\begin{array}{c|c} A + BF & BZ^{-1/2} \\ \hline F & Z^{-1/2} \\ C + DF & DZ^{-1/2} \end{array} \right] \in \mathcal{RH}_\infty$$

where

$$Z = R + B^*XB$$

$$F = -Z^{-1}(B^*XA + D^*C)$$

and $X = X^* \geq 0$ is the unique stabilizing solution

$$A_n^*X(I + BR^{-1}B^*X)^{-1}A_n - X + C^*\tilde{R}^{-1}C = 0$$

where $A_n := A - BR^{-1}D^*C$.

- (b) Suppose that (C, A) is detectable and that (A, B) has no uncontrollable modes on the imaginary axis. Then there is a normalized left coprime factorization $G = \tilde{M}^{-1}\tilde{N}$

$$\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} := \left[\begin{array}{c|c} A + LC & L \\ \hline \tilde{Z}^{-1/2}C & \tilde{Z}^{-1/2}B + \tilde{Z}^{-1/2}D \end{array} \right]$$

where

$$\tilde{Z} = \tilde{R} + CYC^*$$

$$L = -(BD^* + AY C^*)\tilde{Z}^{-1}$$

and $Y = Y^* \geq 0$ is the unique stabilizing solution

$$\tilde{A}_n Y (I + C^* \tilde{R}^{-1} C Y)^{-1} \tilde{A}_n^* - Y + B R^{-1} B^* = 0$$

where $\tilde{A}_n := A - B D^* \tilde{R}^{-1} C$.

- (c) The controllability Gramian P and observability Gramian Q of $\begin{bmatrix} M \\ N \end{bmatrix}$ are given by

$$P = (I + YX)^{-1}Y, \quad Q = X$$

while the controllability Gramian $\tilde{P}$ and observability Gramian $\tilde{Q}$ of $\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix}$ are given by

$$\tilde{P} = Y, \quad \tilde{Q} = (I + XY)^{-1}X.$$

21.4.3 Spectral Factorizations

The following theorem gives a solution to a special class of spectral factorization problems.

Theorem 21.22 Assume $G(z) := \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty$ and $\gamma > \|G(z)\|_\infty$. Then, there exists a transfer matrix $M \in \mathcal{RH}_\infty$ such that $M^*M = \gamma^2 I - G^*G$ and $M^{-1} \in \mathcal{RH}_\infty$. A particular realization of M is

$$M(z) = \left[\begin{array}{c|c} A & B \\ \hline -R^{1/2}F & R^{1/2} \end{array} \right]$$

where

$$R_D = \gamma^2 I - D^* D$$

$$R = R_D - B^* X B$$

$$F = (R_D - B^* X B)^{-1} (B^* X A + D^* C)$$

and $X = X^* \geq 0$ is the stabilizing solution of

$$A_s^* X (I - B R_D^{-1} B^* X)^{-1} A_s - X + C^* (I + D R_D^{-1} D^*) C = 0$$

where $A_s := A + B R_D^{-1} D^* C$.

Similar to the continuous time, we have the following theorem.

Theorem 21.23 Let $G(z) := \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty$ with D full row rank and $G(e^{j\theta})G^*(e^{j\theta}) > 0$ for all θ . Then, there exists a transfer matrix $M \in \mathcal{RH}_\infty$ such that $M^* M = G G^*$. A particular realization of M is

$$M(z) = \left[\begin{array}{c|c} A & B_W \\ \hline C_W & D_W \end{array} \right]$$

where

$$\begin{aligned} B_W &= A P C^* + B D^* \\ D_W^* D_W &= D D^* \\ C_W &= D_W (D D^*)^{-1} (C - B_W^* X A) \end{aligned}$$

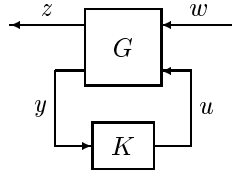
and

$$A P A^* - P + B B^* = 0$$

$$A^* X A - X + (C - B_W^* X A)^* (D D^*)^{-1} (C - B_W^* X A) = 0.$$

21.5 Discrete Time $\mathcal{H}_2$ Control

Consider the system described by the block diagram



The realization of the transfer matrix G is taken to be of the form

$$G(z) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & 0 \end{array} \right] =: \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right].$$

The following assumptions are made:

(A1) (A, B_2) is stabilizable and (C_2, A) is detectable;

(A2) D_{12} is full column rank with $\begin{bmatrix} D_{12} & D_{\perp} \end{bmatrix}$ unitary and D_{21} is full row rank with $\begin{bmatrix} D_{21} \\ \tilde{D}_{\perp} \end{bmatrix}$ unitary;

(A3) $\begin{bmatrix} A - e^{j\theta} I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all $\theta \in [0, 2\pi]$;

(A4) $\begin{bmatrix} A - e^{j\theta} I & B_1 \\ C_2 & D_{21} \end{bmatrix}$ has full row rank for all $\theta \in [0, 2\pi]$.

The problem in this section is to find an admissible controller K which minimizes $\|T_{zw}\|_2$.

Denote

$$A_x := A - B_2 D_{12}^* C_1, \quad A_y := A - B_1 D_{21}^* C_2.$$

Let $X_2 \geq 0$ and $Y_2 \geq 0$ be the stabilizing solutions to the following Riccati equations:

$$A_x^*(I + X_2 B_2 B_2^*)^{-1} X_2 A_x - X_2 + C_1^* D_{\perp} D_{\perp}^* C_1 = 0$$

and

$$A_y(I + Y_2 C_2^* C_2)^{-1} Y_2 A_y^* - Y_2 + B_1 \tilde{D}_{\perp}^* \tilde{D}_{\perp} B_1^* = 0.$$

Note that the stabilizing solutions exist by the assumptions (A3) and (A4). Note also that if A_x and A_y are nonsingular, the solutions can be obtained through the following two symplectic matrices:

$$H_2 := \begin{bmatrix} A_x + B_2 B_2^* (A_x^*)^{-1} C_1^* D_{\perp} D_{\perp}^* C_1 & -B_2 B_2^* (A_x^*)^{-1} \\ -(A_x^*)^{-1} C_1^* D_{\perp} D_{\perp}^* C_1 & (A_x^*)^{-1} \end{bmatrix}$$

$$J_2 := \begin{bmatrix} A_y^* + C_2^* C_2 A_y^{-1} B_1 \tilde{D}_{\perp}^* \tilde{D}_{\perp} B_1^* & -C_2^* C_2 A_y^{-1} \\ -A_y^{-1} B_1 \tilde{D}_{\perp}^* \tilde{D}_{\perp} B_1^* & A_y^{-1} \end{bmatrix}.$$

Define

$$\begin{aligned}
 R_b &:= I + B_2^* X_2 B_2 \\
 F_2 &:= -(I + B_2^* X_2 B_2)^{-1} (B_2^* X_2 A + D_{12}^* C_1) \\
 F_0 &:= -(I + B_2^* X_2 B_2)^{-1} (B_2^* X_2 B_1 + D_{12}^* D_{11}) \\
 L_2 &:= -(A Y_2 C_2^* + B_1 D_{21}^*) (I + C_2 Y_2 C_2^*)^{-1} \\
 L_0 &:= (F_2 Y_2 C_2^* + F_0 D_{21}^*) (I + C_2 Y_2 C_2^*)^{-1}
 \end{aligned}$$

and

$$\begin{aligned}
 A_{F_2} &:= A + B_2 F_2, \quad C_{1F_2} := C_1 + D_{12} F_2 \\
 A_{L_2} &:= A + L_2 C_2, \quad B_{1L_2} := B_1 + L_2 D_{21} \\
 \hat{A}_2 &:= A + B_2 F_2 + L_2 C_2 \\
 G_c(z) &:= \left[\begin{array}{c|c} A_{F_2} & B_1 + B_2 F_0 \\ \hline C_{1F_2} & D_{11} + D_{12} F_0 \end{array} \right] \\
 G_f(z) &:= \left[\begin{array}{c|c} A_{L_2} & B_{1L_2} \\ \hline R_b^{1/2} (L_0 C_2 - F_2) & R_b^{1/2} (L_0 D_{21} - F_0) \end{array} \right].
 \end{aligned}$$

Theorem 21.24 *The unique optimal controller is*

$$K_{opt}(z) := \left[\begin{array}{c|c} \hat{A}_2 - B_2 L_0 C_2 & -(L_2 - B_2 L_0) \\ \hline F_2 - L_0 C_2 & L_0 \end{array} \right].$$

Moreover, $\min \|T_{zw}\|_2^2 = \|G_c\|_2^2 + \|G_f\|_2^2$.

Remark 21.6 Note that for a discrete time transfer matrix $G(z) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_2$, its $\mathcal{H}_2$ norm can be computed as

$$\|G(z)\|_2^2 = \text{Trace}\{D^* D + B^* L_o B\} = \text{Trace}\{D D^* + C L_c C^*\}$$

where L_c and L_o are the controllability and observability Gramians

$$A L_c A^* - L_c + B B^* = 0$$

$$A^* L_o A - L_o + C^* C = 0.$$

Using the above formula, we can compute $\min \|T_{zw}\|_2^2$ by noting that X_2 and Y_2 satisfy the equations

$$A_{F_2}^* X_2 A_{F_2} - X_2 + C_{1F_2}^* C_{1F_2} = 0$$

$$A_{L_2} Y_2 A_{L_2}^* - Y_2 + B_{1L_2} B_{1L_2}^* = 0.$$

For example,

$$\|G_c\|_2^2 = \text{Trace}\{(D_{11} + D_{12}F_0)^*(D_{11} + D_{12}F_0) + (B_1 + D_2F_0)X_2(B_1 + D_2F_0)\}$$

and

$$\|G_f\|_2^2 = \text{Trace } R_b \{(L_0D_{21} - F_0)(L_0D_{21} - F_0)^* + (L_0C_2 - F_2)Y_2(L_0C_2 - F_2)^*\}.$$

♡

Proof. Let x denote the states of the system G . Then the system can be written as

$$\dot{x} = Ax + B_1w + B_2u \quad (21.24)$$

$$z = C_1x + D_{11}w + D_{12}u \quad (21.25)$$

$$y = C_2x + D_{21}w. \quad (21.26)$$

Define $\nu := u - F_2x - F_0w$; then the transfer function from w, ν to z becomes

$$z = \left[\begin{array}{c|cc} A_{F_2} & B_1 + B_2F_0 & B_2 \\ \hline C_{1F_2} & D_{11} + D_{12}F_0 & D_{12} \end{array} \right] \begin{bmatrix} w \\ \nu \end{bmatrix} = G_c w + UR_b^{1/2} \nu$$

where

$$U(s) := \left[\begin{array}{c|c} A_{F_2} & B_2R_b^{-1/2} \\ \hline C_{1F_2} & D_{12}R_b^{-1/2} \end{array} \right].$$

It is easy to shown that U is an inner and that $U \sim G_c \in \mathcal{RH}_2^\perp$. Now denote the transfer function from w to ν by $T_{\nu w}$. Then

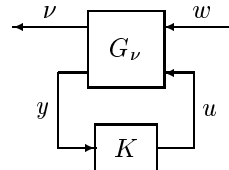
$$T_{zw} = G_c + UR_b^{1/2}T_{\nu w}$$

and

$$\|T_{zw}\|_2^2 = \|G_c\|_2^2 + \|R_b^{1/2}T_{\nu w}\|_2^2 \geq \|G_c\|_2^2$$

for any given stabilizing controller K . Hence if the states (x) and the disturbance (w) are both available for feedback (i.e., full information control) and $u = F_2x + F_0w$, then $T_{\nu w} = 0$ and $\|T_{zw}\|_2 = \|G_c\|_2$. Therefore, $u = F_2x + F_0w$ is an optimal full information control law. Note that

$$\nu = T_{\nu w}w, \quad T_{\nu w} = \mathcal{F}_\ell(G_\nu, K)$$



$$G_\nu = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline -F_2 & -F_0 & I \\ C_2 & D_{21} & 0 \end{array} \right]$$

Since $A - B_2(-F_2) = A + B_2F_2$ is stable, from the relationship between an output estimation (OE) problem and a full control (FC) problem, all admissible controllers for G_ν (hence for the output feedback problem) can be written as

$$K = \mathcal{F}_\ell(M_t, K_{FC}), \quad M_t = \left[\begin{array}{c|c|c} A + B_2F_2 & 0 & \begin{bmatrix} I & -B_2 \end{bmatrix} \\ \hline -F_2 & 0 & \begin{bmatrix} 0 & I \end{bmatrix} \\ \hline C_2 & I & \begin{bmatrix} 0 & 0 \end{bmatrix} \end{array} \right]$$

where K_{FC} is an internally stabilizing controller for the following system:

$$\hat{G}_\nu = \left[\begin{array}{c|c|c} A & B_1 & \begin{bmatrix} I & 0 \end{bmatrix} \\ \hline -F_2 & -F_0 & \begin{bmatrix} 0 & I \end{bmatrix} \\ \hline C_2 & D_{21} & \begin{bmatrix} 0 & 0 \end{bmatrix} \end{array} \right].$$

Furthermore, $T_{\nu w} = \mathcal{F}_\ell(G_\nu, \mathcal{F}_\ell(M_t, K_{FC})) = \mathcal{F}_\ell(\hat{G}_\nu, K_{FC})$. Hence

$$\min_K \|T_{zw}\|_2^2 = \|G_c B_1\|_2^2 + \min_{K_{FC}} \|R_b^{1/2} \mathcal{F}_\ell(\hat{G}_\nu, K_{FC})\|_2^2.$$

Next define

$$\begin{aligned} \tilde{G}_\nu &:= \begin{bmatrix} R_b^{1/2} & 0 \\ 0 & I \end{bmatrix} \hat{G}_\nu \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & R_b^{-1/2} \end{bmatrix} \\ &= \left[\begin{array}{c|c|c} A & B_1 & \begin{bmatrix} I & 0 \end{bmatrix} \\ \hline -R_b^{1/2}F_2 & -R_b^{1/2}F_0 & \begin{bmatrix} 0 & I \end{bmatrix} \\ \hline C_2 & D_{21} & \begin{bmatrix} 0 & 0 \end{bmatrix} \end{array} \right] \end{aligned}$$

and

$$\tilde{K}_{FC} := \begin{bmatrix} I & 0 \\ 0 & R_b^{1/2} \end{bmatrix} K_{FC}.$$

Then it is easy to see that

$$R_b^{1/2} \mathcal{F}_\ell(\hat{G}_\nu, K_{FC}) = \mathcal{F}_\ell(\tilde{G}_\nu, \tilde{K}_{FC})$$

and

$$\min_{K_{FC}} \|R_b^{1/2} \mathcal{F}_\ell(\hat{G}_\nu, K_{FC})\|_2 = \min_{\tilde{K}_{FC}} \|\mathcal{F}_\ell(\tilde{G}_\nu, \tilde{K}_{FC})\|_2.$$

A controller minimizing $\|\mathcal{F}_\ell(\tilde{G}_\nu, \tilde{K}_{FC})\|_2$ is $\tilde{K}_{FC} = \begin{bmatrix} L_2 \\ R_b^{1/2} L_0 \end{bmatrix}$ since the transpose (or dual) of $\mathcal{F}_\ell(\tilde{G}_\nu, \tilde{K}_{FC})$ is a full information feedback problem considered at the beginning of the proof. Hence we get

$$\mathcal{F}_\ell(\tilde{G}_\nu, \begin{bmatrix} L_2 \\ R_b^{1/2} L_0 \end{bmatrix}) = G_f$$

and

$$\min_K \|T_{zw}\|_2^2 = \|G_c\|_2^2 + \|G_f\|_2^2.$$

Finally, the optimal output feedback controller is given by

$$K = \mathcal{F}_\ell(M_t, \begin{bmatrix} L_2 \\ L_0 \end{bmatrix}) = \left[\begin{array}{c|c} \frac{A + B_2 F_2 + L_2 C_2 - B_2 L_0 C_2}{-F_2 + L_0 C_2} & \frac{L_2 - B_2 L_0}{L_0} \end{array} \right].$$

The proof of uniqueness is similar to the continuous time case, and hence omitted. $\square$

It should be noted that in contrast with the continuous time the full information optimal control problem in the discrete time is not a state feedback even when $D_{11} = 0$.

The discrete time $\mathcal{H}_\infty$ control problem is much more involved and it is probably more effective to obtain the discrete solution by using a bilinear transformation.

21.6 Discrete Balanced Model Reduction

In this section, we will show another application of the LFT machinery in discrete balanced model reduction. We will show an elegant proof of the balanced truncation error bound.

Consider a stable discrete time system $G(z)$ and assume that the transfer matrix has the following realization:

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty.$$

Let P and Q be two positive semi-definite symmetric matrices such that

$$APA^* - P + BB^* \leq 0 \quad (21.27)$$

$$A^*QA - Q + C^*C \leq 0. \quad (21.28)$$

Without loss of generality, we shall assume that

$$P = Q = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix}$$

with

$$\begin{aligned}\Sigma_1 &= \text{diag}(\sigma_1 I_{s_1}, \sigma_2 I_{s_2}, \dots, \sigma_r I_{s_r}) \geq 0 \\ \Sigma_2 &= \text{diag}(\sigma_{r+1} I_{s_{r+1}}, \sigma_{r+2} I_{s_{r+2}}, \dots, \sigma_n I_{s_n}) \geq 0\end{aligned}$$

where s_i denotes the multiplicity of σ_i . (Note that the singular values are not necessarily ordered.) Moreover, the realization for $G(z)$ is partitioned conformably with P and Q :

$$G(s) = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array} \right].$$

Theorem 21.25 *Suppose $\Sigma_1 > 0$. Then A_{11} is stable.*

Proof. We shall first assume $\Sigma_2 > 0$. From equation (21.28), we have

$$A_{11}^* \Sigma_1 A_{11} - \Sigma_1 + A_{21}^* \Sigma_2 A_{21} + C_1^* C_1 \leq 0. \quad (21.29)$$

Assume that λ is an eigenvalue of A_{11} ; then there is an $x \neq 0$ such that

$$A_{11}x = \lambda x. \quad (21.30)$$

Now pre-multiply x^* and post-multiply x to equation (21.29) to get

$$(|\lambda|^2 - 1)x^* \Sigma_1 x + x^* A_{21}^* \Sigma_2 A_{21} x + x^* C_1^* C_1 x \leq 0.$$

It is clear that $|\lambda| \leq 1$. However, if $|\lambda| = 1$, say $\lambda = e^{j\theta}$ for some θ , we have

$$A_{21}x = 0, \quad C_1x = 0.$$

These equations together with equation (21.30) imply that

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x \\ 0 \end{bmatrix} = e^{j\theta} \begin{bmatrix} x \\ 0 \end{bmatrix}$$

i.e., $e^{j\theta}$ is an eigenvalue of A . This contradicts the stability assumption of A , so A_{11} is stable.

Now assume that Σ_2 is singular, and we will show that we can remove all those states corresponding to the zero singular values without changing the system stability. For that purpose, we assume $\Sigma_2 = 0$. Then the inequality (21.27) can be written as

$$\begin{bmatrix} A_{11} \Sigma_1 A_{11}^* - \Sigma_1 + B_1 B_1^* & A_{11} \Sigma_1 A_{21}^* + B_1 B_2^* \\ A_{21} \Sigma_1 A_{11}^* + B_2 B_1^* & A_{21} \Sigma_1 A_{21}^* + B_2 B_2^* \end{bmatrix} \leq 0.$$

This implies that

$$A_{11}\Sigma_1 A_{11}^* - \Sigma_1 + B_1 B_1^* \leq 0 \quad (21.31)$$

and

$$A_{21}\Sigma_1 A_{21}^* + B_2 B_2^* \leq 0. \quad (21.32)$$

But inequality (21.32) implies that

$$A_{21} = 0, \quad B_2 = 0.$$

Hence we have

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix},$$

and the stability of A_{11} and A_{22} are ensured.

Substitute this A matrix into the inequality (21.28), and we obtain

$$A_{11}^* \Sigma_1 A_{11} - \Sigma_1 + C_1^* C_1 \leq 0.$$

The subsystem with A_{11} still satisfies inequalities (21.27) and (21.28) with $\Sigma_1 > 0$. This proves that we can assume without loss of generality that $\Sigma_2 > 0$. $\square$

Remark 21.7 It is important to note that the realization for the truncated subsystem

$$G_r = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array} \right]$$

is still *balanced* in some sense¹ since the system parameters satisfy the following equations:

$$A_{11}\Sigma_1 A_{11}^* - \Sigma_1 + A_{12}\Sigma_2 A_{12}^* + B_1 B_1^* \leq 0$$

$$A_{11}^* \Sigma_1 A_{11} - \Sigma_1 + A_{21}^* \Sigma_2 A_{21} + C_1^* C_1 \leq 0.$$

But these equations imply that

$$A_{11}\Sigma_1 A_{11}^* - \Sigma_1 + B_1 B_1^* \leq 0$$

$$A_{11}^* \Sigma_1 A_{11} - \Sigma_1 + C_1^* C_1 \leq 0$$

hold. $\heartsuit$

Theorem 21.26 Suppose $G_r = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array} \right]$. Then $\|G - G_r\|_\infty \leq 2 \sum_{i=r+1}^n \sigma_i$.

In particular, $\|G\|_\infty \leq \|D\| + 2 \sum_{i=1}^n \sigma_i$.

¹Balanced in the sense that the same inequalities as (21.27) and (21.28) are satisfied.

Without loss of generality, we shall assume $\sigma_n = 1$. We will prove that for $\Sigma_2 = \sigma_n I = I$, we have

$$\|G - G_r\|_\infty \leq 2, \quad r = n - 1.$$

Then the theorem follows immediately by scaling and recursively applying this result since the reduced system G_r is still balanced.

It will be seen that it is more convenient to set $\Lambda = \Sigma_1^{1/2}$. The proof of the theorem will follow from the following two lemmas and the bounded real lemma which establishes the relationship between the $\mathcal{H}_\infty$ norm of a transfer matrix and its realizations. (Note that in the following, a constant matrix X is said to be *contractive* or a *contraction* if $\|X\| \leq 1$ and *strictly contractive* if $\|X\| < 1$).

The lemma below shows that for any stable system there is a realization such that $\begin{bmatrix} A & B \end{bmatrix}$ is a contraction, and, similarly, there is another realization such that $\begin{bmatrix} A \\ C \end{bmatrix}$ is a contraction.

Lemma 21.27 *Suppose that a realization of the transfer matrix G satisfies $P = Q = \text{diag}\{\Lambda^2, I\}$; then*

$$\begin{bmatrix} \Lambda^{-1}A_{12} & \Lambda^{-1}A_{11}\Lambda & \Lambda^{-1}B_1 \\ A_{22}\Lambda & A_{21} & B_2 \end{bmatrix}$$

and

$$\begin{bmatrix} A_{21}\Lambda^{-1} & A_{22} \\ \Lambda A_{11}\Lambda^{-1} & \Lambda A_{12} \\ C_1\Lambda^{-1} & C_2 \end{bmatrix}$$

are contractive.

Proof. Since $P = \begin{bmatrix} \Lambda^2 & 0 \\ 0 & I \end{bmatrix}$ satisfies the inequality (21.27),

$$\begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} P & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} A^* \\ B^* \end{bmatrix} \leq P$$

i.e., $\begin{bmatrix} P^{-1/2}AP^{1/2} & P^{-1/2}B \end{bmatrix}$ is a contraction. But

$$\begin{bmatrix} P^{-1/2}AP^{1/2} & P^{-1/2}B \end{bmatrix} = \begin{bmatrix} \Lambda^{-1}A_{12} & \Lambda^{-1}A_{11}\Lambda & \Lambda^{-1}B_1 \\ A_{22}\Lambda & A_{21} & B_2 \end{bmatrix} \begin{bmatrix} 0 & I & 0 \\ I & 0 & 0 \\ 0 & 0 & I \end{bmatrix}.$$

Hence

$$\begin{bmatrix} \Lambda^{-1}A_{12} & \Lambda^{-1}A_{11}\Lambda & \Lambda^{-1}B_1 \\ A_{22}\Lambda & A_{21} & B_2 \end{bmatrix}$$

is also a contraction. The other part follows by a similar argument. $\square$

Lemma 21.28 *Suppose that $X = \begin{bmatrix} X_{11} & X_{12} \\ Z & X_{22} \end{bmatrix}$ and $Y = \begin{bmatrix} Y_{11} & Z \\ Y_{21} & Y_{22} \end{bmatrix}$ are contractive (strictly contractive). Then*

$$M \triangleq \begin{bmatrix} 0 & \frac{1}{\sqrt{2}}X_{11} & X_{12} \\ \frac{1}{\sqrt{2}}Y_{11} & Z & \frac{1}{\sqrt{2}}X_{22} \\ Y_{21} & \frac{1}{\sqrt{2}}Y_{22} & 0 \end{bmatrix}$$

is also contractive (strictly contractive).

Proof. Dilate M to the following matrix:

$$M_d \triangleq \left[\begin{array}{ccc|c} 0 & \frac{1}{\sqrt{2}}X_{11} & X_{12} & \frac{1}{\sqrt{2}}X_{11} \\ \frac{1}{\sqrt{2}}Y_{11} & Z & \frac{1}{\sqrt{2}}X_{22} & 0 \\ Y_{21} & \frac{1}{\sqrt{2}}Y_{22} & 0 & -\frac{1}{\sqrt{2}}Y_{22} \\ \hline \frac{1}{\sqrt{2}}Y_{11} & 0 & -\frac{1}{\sqrt{2}}X_{22} & -Z \end{array} \right].$$

Considering X and Y are contractive, we can easily verify that $M_d^* M_d \leq I$, i.e., M_d is a contraction. $\square$

We can now prove the theorem.

Proof of Theorem 21.26. Note that

$$G_r = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array} \right] = \left[\begin{array}{cc|c} A_{11} & 0 & B_1 \\ 0 & 0 & 0 \\ \hline C_1 & 0 & D \end{array} \right].$$

Hence

$$\frac{1}{2}(G - G_r) = \left[\begin{array}{cccc|c} A_{11} & 0 & 0 & 0 & \frac{1}{\sqrt{2}}B_1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & A_{11} & A_{12} & \frac{1}{\sqrt{2}}B_1 \\ 0 & 0 & A_{21} & A_{22} & \frac{1}{\sqrt{2}}B_2 \\ \hline -\frac{1}{\sqrt{2}}C_1 & 0 & \frac{1}{\sqrt{2}}C_1 & \frac{1}{\sqrt{2}}C_2 & 0 \end{array} \right].$$

Now apply the similarity transformation

$$T = \begin{bmatrix} -\Lambda & 0 & \Lambda & 0 \\ 0 & -I & 0 & I \\ \Lambda^{-1} & 0 & \Lambda^{-1} & 0 \\ 0 & I & 0 & I \end{bmatrix}$$

to the realization of $\frac{1}{2}(G - G_r)$ to get

$$\frac{1}{2}(G - G_r) = \left[\begin{array}{cccc|c} \Lambda A_{11} \Lambda^{-1} & \frac{1}{2} \Lambda A_{12} & 0 & \frac{1}{2} \Lambda A_{12} & 0 \\ \frac{1}{2} A_{21} \Lambda^{-1} & \frac{1}{2} A_{22} & \frac{1}{2} A_{21} \Lambda & \frac{1}{2} A_{22} & \frac{1}{2} B_2 \\ 0 & \frac{1}{2} \Lambda^{-1} A_{12} & \Lambda^{-1} A_{11} \Lambda & \frac{1}{2} \Lambda^{-1} A_{12} & \Lambda^{-1} B_1 \\ \frac{1}{2} A_{21} \Lambda^{-1} & \frac{1}{2} A_{22} & \frac{1}{2} A_{21} \Lambda & \frac{1}{2} A_{22} & \frac{1}{2} B_2 \\ \hline C_1 \Lambda^{-1} & \frac{1}{2} C_2 & 0 & \frac{1}{2} C_2 & 0 \end{array} \right].$$

It is easy to verify that *as a constant matrix* the right hand side of the above realization for $\frac{1}{2}(G - G_r)$ can be written as

$$\left[\begin{array}{cccc} 0 & 0 & I & 0 \\ 0 & \frac{1}{\sqrt{2}} I & 0 & 0 \\ I & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} I & 0 & 0 \\ 0 & 0 & 0 & I \end{array} \right] \hat{M}_d \left[\begin{array}{ccccc} I & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} I & 0 & \frac{1}{\sqrt{2}} I & 0 \\ 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & I \end{array} \right] \quad (21.33)$$

where

$$\hat{M}_d \triangleq \left[\begin{array}{cccc} 0 & \frac{1}{\sqrt{2}} \Lambda^{-1} A_{12} & \Lambda^{-1} A_{11} \Lambda & \Lambda^{-1} B_1 \\ \frac{1}{\sqrt{2}} A_{21} \Lambda^{-1} & A_{22} & \frac{1}{\sqrt{2}} A_{21} \Lambda & \frac{1}{\sqrt{2}} B_2 \\ \Lambda A_{11} \Lambda^{-1} & \frac{1}{\sqrt{2}} \Lambda A_{12} & 0 & 0 \\ C_1 \Lambda^{-1} & \frac{1}{\sqrt{2}} C_2 & 0 & 0 \end{array} \right]. \quad (21.34)$$

According to Theorem 21.12 (a) and (g), the theorem follows if we can show that the realization for $\frac{1}{2}(G - G_r)$ as a constant matrix is a contraction. However, this is guaranteed if $\hat{M}_d$ is a contraction since both the right and left hand matrices in (21.33) are contractive.

Finally, the contractiveness of $\hat{M}_d$ follows from Lemmas 21.27 and 21.28 by identifying

$$Z = A_{22}, \quad X_{11} = \frac{1}{\sqrt{2}} \Lambda^{-1} A_{12}, \quad Y_{11} = \frac{1}{\sqrt{2}} A_{21} \Lambda^{-1}$$

and

$$\left[\begin{array}{c} X_{12} \\ X_{22} \end{array} \right] = \left[\begin{array}{cc} \Lambda^{-1} A_{11} \Lambda & \Lambda^{-1} B_1 \\ \frac{1}{\sqrt{2}} A_{21} \Lambda & \frac{1}{\sqrt{2}} B_2 \end{array} \right] \left[\begin{array}{cc} Y_{21} & Y_{22} \end{array} \right] = \left[\begin{array}{cc} \Lambda A_{11} \Lambda^{-1} & \frac{1}{\sqrt{2}} \Lambda A_{12} \\ C_1 \Lambda^{-1} & \frac{1}{\sqrt{2}} C_2 \end{array} \right].$$

□

21.7 Model Reduction Using Coprime Factors

In this section, we consider lower order controller design using coprime factor reduction. We shall only consider the special case where the normalized right coprime factors are used.

Suppose that a dynamic system is given by

$$G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$$

and assume that the realization is stabilizable and detectable. Recall from Theorem 21.21 that there exists a normalized right coprime factorization $G = NM^{-1}$ such that $\begin{bmatrix} M \\ N \end{bmatrix}$ is inner.

Lemma 21.29 *Let $\nu_i = \sqrt{\lambda_i(YX)}$. Then the Hankel singular values of $\begin{bmatrix} M \\ N \end{bmatrix}$ are given by*

$$\sigma_i = \frac{\nu_i}{\sqrt{1 + \nu_i^2}} < 1.$$

Proof. This is obvious since

$$\sigma_i^2 = \lambda_i(PQ) = \frac{\lambda_i(YX)}{1 + \lambda_i(YX)}$$

and $\lambda_i(YX) \geq 0$. □

It is known that there exists a transformation such that X and Y are balanced:

$$X = Y = \Pi = \begin{bmatrix} \Pi_1 & 0 \\ 0 & \Pi_2 \end{bmatrix}$$

with $\Pi_1 = \text{diag}[\nu_1 I_{s_1}, \dots, \nu_r I_{s_r}] > 0$.

Now partitioning the system G and matrix F accordingly,

$$G = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array} \right] \quad F = \begin{bmatrix} F_1 & F_2 \end{bmatrix}.$$

Then the reduced coprime factors

$$\begin{bmatrix} \hat{M} \\ \hat{N} \end{bmatrix} := \left[\begin{array}{c|c} \frac{A_{11} + B_1 F_1}{F_1} & \frac{B_1 Z^{-1/2}}{Z^{-1/2}} \\ \hline C_1 + D F_1 & D Z^{-1/2} \end{array} \right] \in \mathcal{RH}_\infty$$

satisfy the following error bound.

Lemma 21.30 $\left\| \begin{bmatrix} M \\ N \end{bmatrix} - \begin{bmatrix} \hat{M} \\ \hat{N} \end{bmatrix} \right\|_\infty \leq 2 \sum_{i \geq r+1} \sigma_i = 2 \sum_{i \geq r+1} \frac{\nu_i}{\sqrt{1 + \nu_i^2}}.$

Proof. Analogous to the continuous time case. $\square$

Remark 21.8 It should be understood that the reduced model can be obtained by directly computing X and P and by obtaining a balanced model without solving the Riccati equation for Y . $\heartsuit$

This reduced coprime factors combined with the robust or $\mathcal{H}_\infty$ controller design methods can be used to design lower order controllers so that the system is robustly stable and some specified performance criteria are satisfied. We will leave the readers to explore the utility of this model reduction method. However, we would like to point out that unlike the continuous time case, the reduced coprime factors in discrete time *may not* be normalized. In fact, we can prove a more general result.

Lemma 21.31 *Let a realization for $N(z) \in \mathcal{RH}_\infty$ be given by*

$$N(z) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$$

with A stable. Suppose that there exists an $X = X^ \geq 0$ such that*

$$\left[\begin{array}{cc} A^*XA - X + C^*C & A^*XB + C^*D \\ B^*XA + D^*C & B^*XB + D^*D - I \end{array} \right] = 0. \quad (21.35)$$

Then N is an inner, i.e., $N \sim N = I$. Moreover, if the realization for

$$N = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array} \right]$$

is also balanced with

$$X = \Sigma = \left[\begin{array}{cc} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{array} \right]$$

and

$$A\Sigma A^* - \Sigma + BB^* = 0$$

with $\Sigma_1 > 0$, then the truncated system

$$N_r = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array} \right]$$

is stable and contractive, i.e., $N_r \sim N_r \leq I$.

Proof. Pre-multiply equation (21.35) by

$$U = \begin{bmatrix} \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} & 0 \\ 0 & I \end{bmatrix}$$

and post-multiply equation (21.35) by U^* to get

$$\begin{bmatrix} A_{11}^* \Sigma_1 A_{11} - \Sigma_1 + C_1^* C_1 + A_{21}^* \Sigma_2 A_{21} & A_{11}^* \Sigma_1 B_1 + C_1^* D + A_{21}^* \Sigma_2 B_2 \\ B_1^* \Sigma_1 A_{11} + D^* C_1 + B_2^* \Sigma_2 A_{21} & B_1^* \Sigma_1 B_1 + D^* D - I + B_2^* \Sigma_2 B_2 \end{bmatrix} = 0$$

or

$$\begin{bmatrix} A_{11}^* \Sigma_1 A_{11} - \Sigma_1 + C_1^* C_1 & A_{11}^* \Sigma_1 B_1 + C_1^* D \\ B_1^* \Sigma_1 A_{11} + D^* C_1 & B_1^* \Sigma_1 B_1 + D^* D - I \end{bmatrix} = - \begin{bmatrix} A_{21}^* \\ B_2^* \end{bmatrix} \Sigma_2 \begin{bmatrix} A_{21}^* \\ B_2^* \end{bmatrix}^*.$$

This gives

$$\begin{bmatrix} A_{11} & B_1 \\ C_1 & D \end{bmatrix}^* \begin{bmatrix} \Sigma_1 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} A_{11} & B_1 \\ C_1 & D \end{bmatrix} - \begin{bmatrix} \Sigma_1 & 0 \\ 0 & I \end{bmatrix} \leq 0.$$

Now let $T = \Sigma_1^{1/2}$, and then

$$\left\| \begin{bmatrix} T A_{11} T^{-1} & T B_1 \\ C_1 T^{-1} & D \end{bmatrix} \right\| \leq 1$$

i.e., $\|N_r\|_\infty \leq 1$. It is clear that N_r is inner if $\begin{bmatrix} A_{21}^* \\ B_2^* \end{bmatrix} \Sigma_2 = 0$ (although the converse may not be true). $\square$

21.8 Notes and References

The results for the discrete Riccati equation are based mostly on the work of Kucera [1972] and Molinari [1975]. Numerical algorithms for solving discrete ARE with singular A matrix can be found in Arnold and Laub [1984], Dooren [1981], and references therein. The matrix factorizations are obtained by Chu [1988]. The normalized coprime factorizations are obtained by Meyer [1990] and Walker [1990]. The detailed treatment of discrete time $\mathcal{H}_\infty$ control can be found in Stoorvogel [1990], Limebeer, Green, and Walker [1989], and Iglesias and Glover [1991].

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